
Use Cases of Terraform

Terraform is used across DevOps, Cloud, SRE, and Platform Engineering because it can automate infrastructure on **any cloud or platform**.

Below are the major use cases:

1 Provisioning Cloud Infrastructure (Most Common Use Case)

Terraform can automatically create and manage cloud resources like:

- VPCs and Virtual Networks
- Subnets
- EC2/VM instances
- Security Groups / NSGs / Firewalls
- Load Balancers
- Databases (RDS, CosmosDB, Cloud SQL)
- Storage (S3, Blob, GCS)

Example: Deploying a full 3-tier architecture (Web, App, DB) using code.

2 Multi-Cloud Deployments

Terraform can manage AWS + Azure + GCP simultaneously in one project.

Examples:

- Backup environments across multiple clouds
- Disaster recovery on another cloud
- Vendor-lock-in avoidance
- Global expansion with multiple providers

This is something tools like CloudFormation or ARM cannot do.

3 Infrastructure Standardization Through Modules

Terraform modules allow companies to create reusable templates.

Example modules:

- Standard VPC module
- Standard Kubernetes cluster module
- Standard S3 bucket module with compliance tags

This ensures:

- Consistency
 - Security
 - Compliance
 - Faster deployments
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4 Kubernetes Infrastructure Provisioning

Terraform can create:

- EKS (AWS Kubernetes)
- AKS (Azure Kubernetes)
- GKE (Google Kubernetes)
- Kubernetes objects using the Kubernetes provider

Examples:

- Automate cluster creation
 - Manage Helm charts
 - Deploy namespaces, configmaps, services
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5 CI/CD Pipeline Infrastructure Automation

Terraform is often integrated with:

- GitHub Actions
- Jenkins
- GitLab CI
- Azure DevOps
- AWS CodePipeline

Use case:

When code is merged → Terraform automatically:

- Creates infra
- Updates infra
- Destroys infra for PR environments

This enables **full DevOps automation**.

6 Network Automation

Terraform can manage advanced networking components:

Examples:

- VPC peering
- Transit gateways
- Route tables
- VPN tunnels
- Load balancers
- DNS (Cloudflare, Route53, Azure DNS)

Large enterprises use Terraform to automate **entire network topologies**.

7 Policy Enforcement & Compliance (with Sentinel / OPA)

Terraform can enforce rules such as:

- No public S3 buckets

- EC2 instances must be `t3.small` or above
- Tags must include CostCenter
- Only encrypted disks allowed

This ensures **security and governance**.

8 Automating Entire Environments (Dev, QA, Staging, Prod)

Using Terraform, you can clone an entire environment with a single command.

Example:

```
terraform apply -var="env=dev"  
terraform apply -var="env=prod"
```

Everything is created with the same baseline.

9 Managing SaaS Services (GitHub, Datadog, Cloudflare)

Terraform is not limited to cloud infra.

Examples:

- Create GitHub repositories
- Create IAM users & groups
- Configure Cloudflare CDN

- Set up Datadog monitoring
 - Configure Vault policies
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Ephemeral / On-Demand Environments (Cost Saving)

Terraform is used to create environments when required and destroy them when the work is done.

Example:

- QA opens a PR → Terraform creates a temporary test environment
- Once merged → Terraform destroys it

This saves 60–80% cloud cost.

Disaster Recovery Automation

Terraform can replicate infrastructure in different regions/clouds.

Example:

- Primary region → us-east-1
- Secondary region → ap-south-1
- Terraform keeps both in sync

Useful for:

- Failover

- Backup
 - Recovery planning
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12 Hybrid Cloud or On-Prem + Cloud Integration

Terraform supports:

- VMware vSphere
- On-prem servers
- Cloud resources

Example:

Using Terraform to:

- Create networks in AWS
 - Extend connectivity to on-prem
 - Provision load balancers & firewalls
 - Create hybrid Kubernetes clusters
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